Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2 years of production cloud infrastructure experience building Terraform, Kubernetes, and CI/CD automation on AWS with Python and Go. Skilled in production debugging, reliability, cost optimization, and automated testing. Current research in LLM-powered Android testing.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Deep Learning.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object-Oriented Programming, Data Mining, Machine Learning.

TECHNICAL SKILLS

Programming: Python, Go, Java, JavaScript, TypeScript, C/C++, SQL.

Web & Backend: Microservices Architecture, REST APIs, FastAPI, Flask, Django, Node.js, React, HTML/CSS.

Databases: PostgreSQL, MySQL, DynamoDB, Redis.

Developer Tools: AWS, Azure, Terraform, Kubernetes, Docker, Ansible, ArgoCD, Helm, GitLab CI/CD, Linux, Bash, Git, Vault.

AI & Data: Gemini, Vertex AI, LangChain, PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Built **Terraform and Ansible** Infrastructure as Code provisioning, standardizing setup across **40+ production environments**.
- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services using CloudWatch and Datadog.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with automated rollback restoring full environments within 15 minutes.
- Reduced monthly AWS spend by **40%** ($15K to $9K) via EC2 rightsizing and automated cleanup of unused resources.
- Built self-service developer platform (dashboard plus CLI/YAML automation) adopted by **15 engineers**, cutting environment start/stop/delete time by up to **90%** and provisioning time by **70%**.
- As Quality Coach and SonarQube Specialist, raised code coverage **35% to 78%**; mentored **5 engineers**.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated AWS provisioning with reusable **Terraform** modules, reducing deployment time by **30%**.
- Developed **Go and Terratest** tests validating AWS provisioning, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, FastAPI, Docker, PostgreSQL, Redis] Jul 2026

- Built Python **FastAPI** service serving real-time demand predictions over **263 zones** with PostgreSQL and Redis caching, containerized with **Docker** for cloud deployment; Poisson hurdle model cut RMSE **43%**.

PolicyPilot | Medical Benefit Policy Intelligence Portal [TS, Node, Docker, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).